A.E.G.I.S.

AUTOMATED EQUITY & GOVERNANCE INSPECTION SYSTEM

The Real-Time Control Layer for Fair AI

AI systems are already making high-stakes decisions. A.E.G.I.S. makes those decisions measurable, explainable, and correctable in real time.

TEAM: QUANTUM DRIFT

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THE BLIND SPOT

AI Audits Fail After Deployment

Most bias tools stop at one-time evaluation.

They may detect unfairness once, but they do
not help teams trace hidden proxies, understand individual harm, or respond when fairness degrades in production.

THE RESULT

Unfair outcomes, weak accountability, and high compliance risk.

What is broken today:

- Static fairness checks miss what happens **after deployment**
- Proxy features can **preserve bias** even when sensitive columns are removed
- Teams get a bias report, but not a **practical mitigation workflow**
- There is **no live alerting** when model behavior shifts over time

THE DIFFERENTIATOR

A.E.G.I.S. turns fairness from a report into a control system

Most tools answer one question:

"Is this model biased?"

A.E.G.I.S. answers four:

1. Where is the bias coming from?
2. Who is being affected?
3. What mitigation reduces it fastest?
4. Is the model drifting back into unfairness?

Core capabilities:

Proxy Bias Radar
surfaces risky feature relationships

Decision Explainer
shows why a prediction changed

Mitigation Engine
applies fairness-aware threshold optimization

Sentinel Monitor
detects fairness drift on live predictions

Detect. Explain. Reduce. Monitor. All in one workflow.

UNDER THE HOOD

Technical Architecture

1	**Discovery Layer**	Correlation, mutual information, and subgroup analysis identify hidden proxies and imbalanced behavior.
2	**Diagnosis Layer**	Fairness metrics compare outcomes across protected groups, including demographic parity and equal opportunity.
3	**Explanation Layer**	SHAP-based interpretation and counterfactual checks explain why a decision was made and how it could flip.
4	**Mitigation Layer**	Fairlearn postprocessing applies group-specific threshold optimization to reduce unfairness while preserving model performance.
5	**Monitoring Layer**	A live pipeline watches incoming predictions, tracks subgroup shifts, and triggers fairness alerts when thresholds are crossed.

Tech Stack: Python · Streamlit · Scikit-learn · Fairlearn · SHAP · Pandas · NumPy

THE PROOF

Live Demo: From Bias Detection to Bias Control

Demo flow:

- Upload a real dataset
- Detect subgroup unfairness
- Explain one harmful prediction
- Apply mitigation
- **Compare before vs after**
- **Trigger live drift alert**

Before / After Results

METRIC	BEFORE A.E.G.I.S.	AFTER A.E.G.I.S.
Bias Gap	**34%**	**1.8%**
Accuracy	85.0%	83.8%

⚠ **Live Sentinel Alert:**
Bias Drift Detected — approval rate for Group B fell below fairness threshold.

WHAT THIS PROVES: A.E.G.I.S. does not just identify unfairness. It shows a measurable path to reduce it and keep watching it after deployment.

WHY THIS STANDS OUT

Other teams may show that AI bias exists.

A.E.G.I.S. shows how to operationalize fairness as an active engineering workflow.

We are not building another fairness dashboard.

We are building the control layer that AI systems need before they can be trusted at scale.